PRACTICAL – 4

JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Aim: To enhance a StudentHub website using JavaScript by implementing DOM manipulation, event handling, collapsible FAQ, notification banner, hamburger menu, and light/dark theme switching using localStorage.

Implementation

1. DOM Selection

JavaScript is used to select HTML elements from the webpage.

Example:

let themeBtn = document.getElementById("themeBtn");

The getElementById() method selects an HTML element using its ID.

Another example:

let questions = document.querySelectorAll(".question");

This selects all elements having the class question.

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2. Event Handling

Event listeners are used to perform an action when the user interacts with the webpage.

Example:

```
themeBtn.addEventListener("click", function() {
    document.body.classList.toggle("dark");
});
```

When the user clicks the button, the dark class is added or removed.

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3. Light/Dark Theme

A light/dark theme switcher was implemented using JavaScript and CSS.

document.body.classList.toggle("dark");

The classList.toggle() method adds the dark class if it is not present and removes it if it is already present.

The selected theme is stored using:

localStorage.setItem("theme", "dark");

The saved theme is restored when the page is loaded.

```
if (localStorage.getItem("theme") === "dark") {
    document.body.classList.add("dark");
}
```

---

4. Collapsible FAQ

The FAQ section allows users to click on a question to show or hide its answer.

```
question.addEventListener("click", function() {
    let answer = this.nextElementSibling;

    if (answer.style.display === "block") {
        answer.style.display = "none";
    } else {
        answer.style.display = "block";
    }
});
```

This provides a simple interactive FAQ section without loading another page.

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5. Notification Banner

A notification banner is displayed at the top of the webpage.

The close button hides the notification when clicked.

```
closeBtn.addEventListener("click", function() {
    notification.style.display = "none";
});
```

This demonstrates event handling and DOM style manipulation.

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6. Hamburger Menu

A hamburger menu is used for smaller screen sizes.

```
menuBtn.addEventListener("click", function() {
    nav.classList.toggle("show");
});
```

When the menu button is clicked, the navigation menu is shown or hidden.

This improves usability on mobile devices.

Code:-

```
let themeButton = document.createElement("button");

themeButton.innerText = "Dark Mode";

themeButton.style.margin = "10px";

document.querySelector("header").appendChild(themeButton);

themeButton.addEventListener("click", function () {

    document.body.classList.toggle("dark")

    if (document.body.classList.contains("dark")) {

        localStorage.setItem("theme", "dark");

        themeButton.innerText = "Light Mode";

    } else {

        localStorage.setItem("theme", "light");

        themeButton.innerText = "Dark Mode";

    }



});
```

```
// Remember theme after refresh


if (localStorage.getItem("theme") == "dark") {
    document.body.classList.add("dark");
    themeButton.innerText = "Light Mode";
}
let questions = document.querySelectorAll("article h3");
questions.forEach(function (question) {
    question.addEventListener("click", function () {
        let answer = question.nextElementSibling;
        if (answer.style.display == "none") {
            answer.style.display = "block";
        } else {
            answer.style.display = "none";
        }
    });
});
let notificationButton = document.createElement("button");
notificationButton.innerText = "Show Notification";
notificationButton.style.margin = "10px";
document.querySelector("main").prepend(notificationButton);
```

```
notificationButton.addEventListener("click", function () {
    alert("New StudentHub notification!");
});

let modalButton = document.createElement("button");
modalButton.innerText = "Open Popup";
modalButton.style.margin = "10px";
document.querySelector("main").prepend(modalButton);
let modal = document.createElement("div");
modal.innerHTML = `
    <h2>StudentHub</h2>
    <p>Welcome to StudentHub!</p>
    <button id="closeModal">Close</button
modal.style.display = "none";
modal.style.background = "white";
modal.style.padding = "20px";
modal.style.border = "2px solid black";
document.querySelector("main").prepend(modal);
modalButton.addEventListener("click", function () {
    modal.style.display = "block";
});
document.getElementById("closeModal").addEventListener("click", function
```

```
() {
    modal.style.display = "none";
});
let menuButton = document.createElement("button");
menuButton.innerText = "☰ Menu";
menuButton.style.margin = "10px";
document.querySelector("header").appendChild(menuButton);
let nav = document.querySelector("nav");
menuButton.addEventListener("click", function () {
    if (nav.style.display == "none") {
        nav.style.display = "block";
    } else {
        nav.style.display = "none";
    }
});
let events = document.querySelectorAll("main article");
if (events.length > 0) {
    let current = 0;

    // Hide all events except first
    for (let i = 1; i < events.length; i++) {
        events[i].style.display = "none";
```

```
}
let nextButton = document.createElement("button");
nextButton.innerText = "Next";
let previousButton = document.createElement("button");
previousButton.innerText = "Previous";
document.querySelector("main").appendChild(previousButton);
document.querySelector("main").appendChild(nextButton);
nextButton.addEventListener("click", function () {
    events[current].style.display = "none";
    current++;
    if (current >= events.length) {
        current = 0;
    }
    events[current].style.display = "block";
});
previousButton.addEventListener("click", function () {
    events[current].style.display = "none";
   current--;
    if (current < 0) {
        current = events.length - 1;
    }
    events[current].style.display = "block";
```

```
    });

}
```